



Indian Institute of Technology (Indian School of Mines) Dhanbad

Department of Mining Engineering

Mid Semester Winter Examination 2023-2024 for B.Tech. + Ph.D. Students (36 nos.)

Subject: Deep Coal Mining (Code: MND 412)

Venue: NLHC LH 14-15

Time: 120 minutes (02:00 PM – 04:00 PM)

Date: 02-03-2024

Maximum Marks: 64

Answer questions of all the three sections and marks are assigned against each section

Section A: Short Answer Type Question (Answer in 1-2 lines/pointwise) – 15 marks

Answer all the questions and each question carries 1 mark

1. Churcha and Jhanjra are the only two underground coal mines of India which are making profits. Discuss the reasons.

Fully mechanised working

2. Which are those four Indian states contributing about 80% of the total coal production in the country?

Odisha, Jharkhand, Madhya Pradesh, Chhattisgarh

3. Discuss the statistics related to coal production from underground and opencast mining methods and also the coal import of last 10 years of India.

Over the last decade, coal production in India has increased by a compound annual growth. In India, opencast mining is the predominant way of extracting coal (96.26 % of the total production in the financial year 2022-23). Around 893 Mt of coal was produced in 2022-23 compared to 500 Mt in 2013-14. Around 96% and 4% was the share of opencast and underground mining methods respectively in 2022-23 compared to 91% and 9% was the share of opencast and underground mining methods respectively in 2013-14. Around 240 Mt of coal was imported in 2022-23 compared to 170 Mt in 2013-14.

4. Brief about the crisis of coal and power in India.

Demand is not meeting production. Indian government said that India surpassed 2021's maximum demand of 200.539 GW (July 7, 2021) and that the demand is expected to reach about 215-220 GW in the May-June period. There is a mix of reasons that have led to the coal situation, according to power sector experts: the rising global price of coal, lack of management of stockpiles at thermal power stations, the poor financial health of power distribution companies (DISCOMs), the impact of the Russia-Ukraine war, and a sudden hike in power demand.

5. Discuss the problems of deep coal mining.

High induced stresses, rock mass characterisation, insitu stresses, pillar design, support design, subsidence.

6. What are the problems associated with the Dishergarh Seam of Chinakuri mine of SECL?

Coal bump/ Rock burst

7. Enumerate the strata mechanics problems encountered while extraction of 10.7 m thick No. 3 Coal Seam of GDK No. 8 Incline of SCCL coal using Blasting Gallery Method.

Air blast

8. What is Energy Release Rate (ERR)?

ERR is the energy released by mining a unit area of the orebody or seam. When ore is extracted, it becomes free of stresses, the exact equivalent of which gets transmitted to the surrounding rock, which thus undergoes an increase in stresses. Salamon (1984) has shown that $ERR \propto \frac{\sigma_1^2(1-\nu^2)h}{2E}$ where σ_1 = major principal stress, E, ν = elastic constants, h = extraction thickness.

9. What are the characteristics of a hard massive roof stratum? Name some Indian mines experiencing the issues and challenges associated with hard massive roof.

Considerable higher value of thickness (> 20 m), compressive strength (> 40 MPa), bridging strata, RMR (> 60), caveability index (> 5000). Churcha.

10. Critically analyse the role of coal and renewables in India's Electricity mix by 2030.

India's COP 26 goals for 2030 include an increase in non-fossil fuel-based energy capacity to 500 GW and reduction of total projected carbon emissions by 1 billion tonnes. In 2020, India used 77% fossil fuels for its electricity generation, of which 72.5% came from coal, 4.2% came from natural gas and 0.3% came from oil. COP26 Summit Glasgow, Nov, 2021- PM Modi promised to improve share of renewables in electricity to 50% by 2030.

11. Discuss the reasons of burial or trapping of Continuous Miner machine in Bord & Pillar Mining Method.

Hidden discontinuities; judicious reduction; overconfidence of continuous miner operator; inefficient breaker line support and design of rib/snook; encroachment during roof fall; overhanging of roof strata

12. How to decide the locations of different geotechnical instruments in a Continuous Miner based mechanised depillaring panel?

Locations of different geotechnical instruments in a Continuous Miner based mechanised depillaring panel are based on the following:

Horizon of key strata

Roof span for major falls

Proposed locations of RBBLS and rib/snook to be created

Width of panel

Manner of pillar extraction

13. What are the rock mechanics principle involved while combating and controlling rock bursts?

The basic concept of destressing or transferring high-stress concentrations from one portion of a mine structure to another is not new for combatting bursts. Fracturing or softening coal or rock to control stress build up has been practised in various mines and has proved helpful (Neyman et al., 1972). In coal mines, destressing the active working face is the most logical method to prevent bumps. Coal and in some instances roof and floor rock, is intentionally fractured and made to fail. As a result no high-stress build up can occur in the fractured zone, and the load is transferred to another part of the mine structures.

14. Discuss the recovery and design issues associated with highwall mining method.

Design of slender web pillars

Avoid premature caving over cutter head of Highwall miner

Prevent Subsidence

Paste filling to maximize pillar extraction

Laboratory studies for pillar strength to consider size effects

Numerical simulation studies

Consumption of cutting picks

15. How to decide the depth of anchorage point and limiting value of roof sagging while installing Auto Warning Tell Tales in a Continuous Miner based mechanised depillaring panel?

Length of roof rock bolts and rock load height

Lithology data about probable horizon of bed separation

Section B: Descriptive Answer Type Question (Answer in 3-5 lines/pointwise) – 24 marks

Answer all the questions and each question carries 2 marks

16. Enumerate the advantages of mechanised working and Longwall technology over semi-mechanised depillaring in Bord and Pillar Mining Method mentioning the name of mechanised equipment involved.

Continuous Miner:

Faster rate of coal production

Lesser strata control issues compared to semi-mechanised

Daily production more than semi-mechanised (300 to 2000 t)

High speed of goaf formation

No danger of spontaneous heating

Less accidents and highly safe

Geotechnical instruments are efficient to predict apprehensive conditions (impending roof fall)

Longwall:

Suitable for extraction at higher depth

Highly productive and safe compared to other mining methods (8,000 t/day)

Good ventilation at the face, minimum air leakage

Full seam extraction in one operation

Good recovery and continuous production

Chances of roof fall, side fall is low as roof is supported

Gassy seam can be worked

Works under weak roof

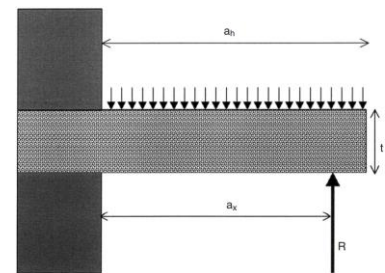
17. Differentiate between (i) main fall and local fall (ii) first goafing and subsequent goafing

Local fall – Roof fall occurring in the goaf immediately just after removing a part (withdrawal of natural support) of coal pillar. Its effect does not reach up to the surface as immediate roof of up to 1-3 m only caves in the goaf.

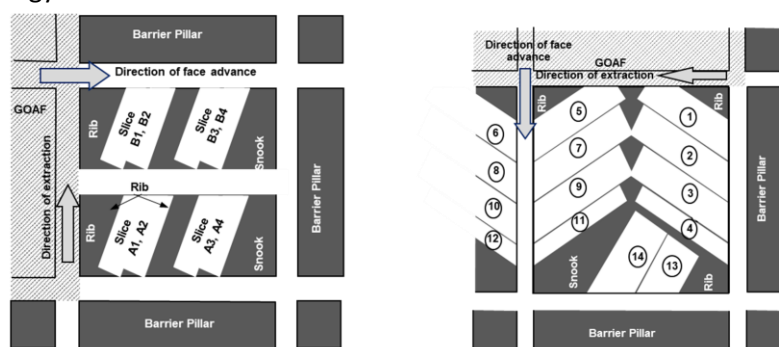
Main fall – It occurs after local fall in the goaf when the length of unsupported roof span in the goaf is equal to width of the panel. Its effect goes up to the surface and causes subsidence. It also occurs when area of extraction (A) is more than square of depth (D) [$A \geq D^2$]

First goafing – It can be also called as main fall. Panel spans required to cause the first goaf to be formed has been found to be equal to panel width. It is the phenomenon of failure of plate roof in the goaf.

Subsequent goafing – once the first goaf has formed, significantly smaller spans (or overhangs) are required for continued overburden layer failure due to conversion of plate (supported from all sides) into cantilever (supported from only one side). Once the first goaf has formed, the horizontal stress is reduced and failure will occur at more frequent intervals. It is shown that bending induced tension is the most likely mode of failure in an unjointed roof layer. In roof layers with a high frequency of weak or uncemented jointing, failure can be expected to occur at spans equal to the joint spacing. In this context, failure will occur at the smallest of the spans required to induce tensile roof failure and the joint spacing.



18. Describe the different adopted manner of pillar extraction using Continuous Miner technology.

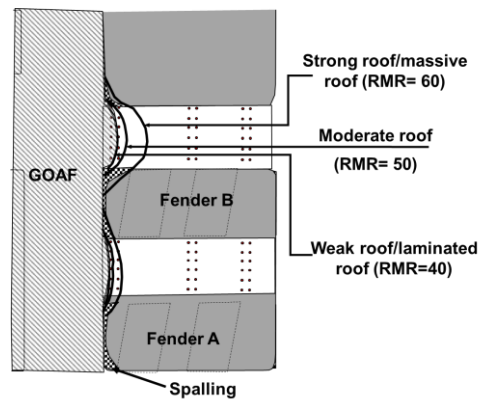


(a) Splitting and slicing/pocket and fender (b) Single pass extraction/Fish-bone/Christmas Tree

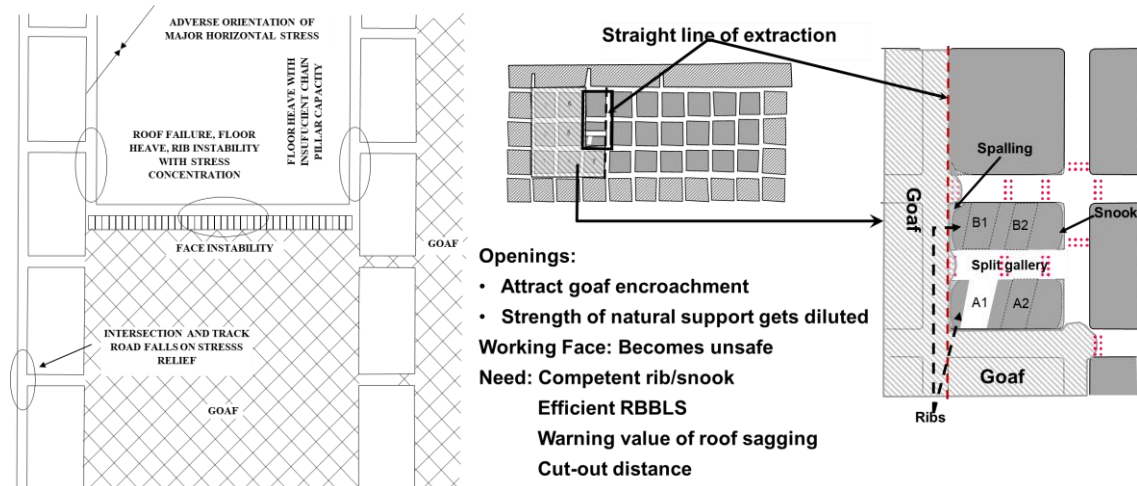
Splitting and slicing requires splitting of a larger pillar (> 30 m x 30 m) at higher depth of cover (> 100 m) and installation of support system including breaker line support in the split gallery and inclined slicing for extraction of the entire pillar from the split and existing original galleries. It is slow.

Single pass extraction does not require splitting of a smaller pillar (< 20 m x 20 m) at shallow depth of cover (< 100 m) and almost a pillar is extracted by left and right slicing from the existing original galleries. It is very fast.

19. Explain the conceptual model for encroachment of different nature of roof in absence of an efficient breaker line support and incompetent rib/snook at the goaf edge.



20. Discuss the strata control issues of working with (i) Continuous Miner and (ii) Longwall technologies.



21. Explain caveability index, abutment angle, bridging strata and bulking factor.

The bulking factor is the ratio or percentage of the volume change of excavated material to the volume of the original in situ volume before excavation.

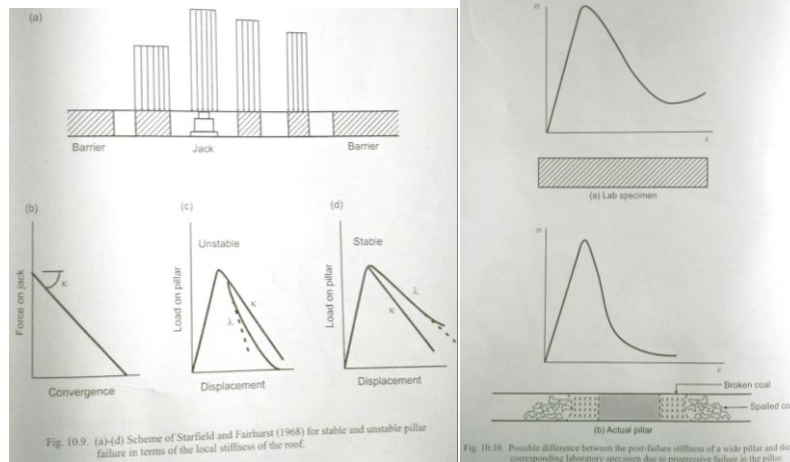
The abutment angle concept is used to calculate the magnitude of abutment loading adjacent to a gob area. It considers an angle between the vertical plane and the panel roof in order to calculate the transferred load to the abutments when the panel is mined

22. Discuss design of yield pillar with respect to stiffness.

The concept of using post-failure pillar behaviour in bord and pillar design, particularly at greater depths, was put forward by Starfield and Fairhurst (1968). Pillar failure in a bord and pillar panel surrounded by wide, strong chain pillars can be catastrophic (unstable) or progressive (stable) depending on strata stiffness and proximity of the chain pillars. The behaviour will thus be akin to that of a rock specimen loaded in a stiff machine. Salamon (1970) has given a complete mathematical treatment for a panel of pillars (which may be of different size). The concept put forward by Starfield and Fairhurst is however much simpler to understand. If a pillar in a panel is replaced by a hydraulic jack as in **Figure 10.9**, which is slowly retracted, the initial force on it due to rock load will drop linearly with a corresponding increase in roof lowering of roof-to-floor convergence, **Figure 10.9b**. The slope of this characteristic, which is termed the local mine stiffness, will be different for different pillar locations, depending on proximity of the stable chain pillars. We now replace the jack by the pillar itself and superpose the local stiffness on the post-failure characteristic of this pillar. If the safety factor of this pillar is less than 1.0 the failure will be stable or unstable depending on whether the local stiffness is steeper or gentler than the steepest part of the post failure stiffness of the pillar, **Figure 10.9c** and **9d**. Thus for stable pillar failure, $\kappa \geq \lambda$

In an array of uniform pillars surrounded by wide chain pillars, the local stiffness will have the greatest negative value next to the chain pillars and the least at the panel centre. Salamon (1970) has shown that pillar failure cannot be stable if a pillar undergoes tributary area load. This means that for the yield pillar method to succeed, panels must necessarily be narrow

along one side but as long as desired along the other side. Also, numerical modelling will be required for estimating the load on pillars as well as the local mine stiffness.



Procedure for Design

The procedure for designing the yield pillar technique for a panel of bord and pillar mining will thus consist of two steps:

- (i) Numerical modelling for pillar load and local mine stiffness.
- (ii) Determination of the realistic post-failure pillar characteristic, whose steepest slope should be less than the local stiffness.

Any suitable numerical method may be used, trying out several panel widths and pillar sizes to ensure that the pillar(s) at the centre of the panel does not undergo the tributary area load and that its safety factor is somewhat less than 1.0.

When these requirements are satisfied, the numerical model has to be run again with the central pillar(s) being removed. The two models will give the roof-to-floor convergence C_p , with the pillar in place, C_e with the pillar removed and σ_z the average vertical stress on the pillar or pillar load. The local mine stiffness is then calculated as $\kappa = \frac{\sigma_z A}{C_e - C_p}$, where A is the

plan area of the pillar. This design method assumes that each pillar offers resistance to the roof and floor at a point and not over an area, as reflected in Equation which involves the mean vertical stress and the mean convergence values. This can at best be acceptable for slender pillars. Since pillar failure proceeds from the outer edges to the pillar centre, each element of the pillar experiences higher horizontal stress as the failure travels inwards. Then again, the horizontal stress falls partly due to the adjoining crushed coal, or wholly, if spalling takes place. Thus, in a wider pillar, a situation shown in Figure 10.10 may arise when the pillar has a steeper post-failure curve than its replica tested in the laboratory. A realistic estimation of the post-failure characteristic of wider pillars ($w/h > 4-6$) becomes rather difficult.

Two procedures can be given for determining the post-failure pillar stiffness in the light of the above discussion:

1. Test replicas of the pillar using the same coal either in a closed-loop servo machine in the laboratory or in situ using servo-controlled multiple jacks as done by Wagner (1974). The second test, though better, is cumbersome and expensive.
2. Numerical modelling of the pillar with strain softening and with realistic inputs of insitu stresses and other properties.

23. How to design chain pillar in Longwall and Bord & Pillar working?

Atypical chain pillar, during its life, will be subjected to the following types of loads:

- (a) Development load before extraction starts.
- (b) Front abutment load, when the face is at the pillar.
- (c) Side abutment load, when the face advances further from the pillar.

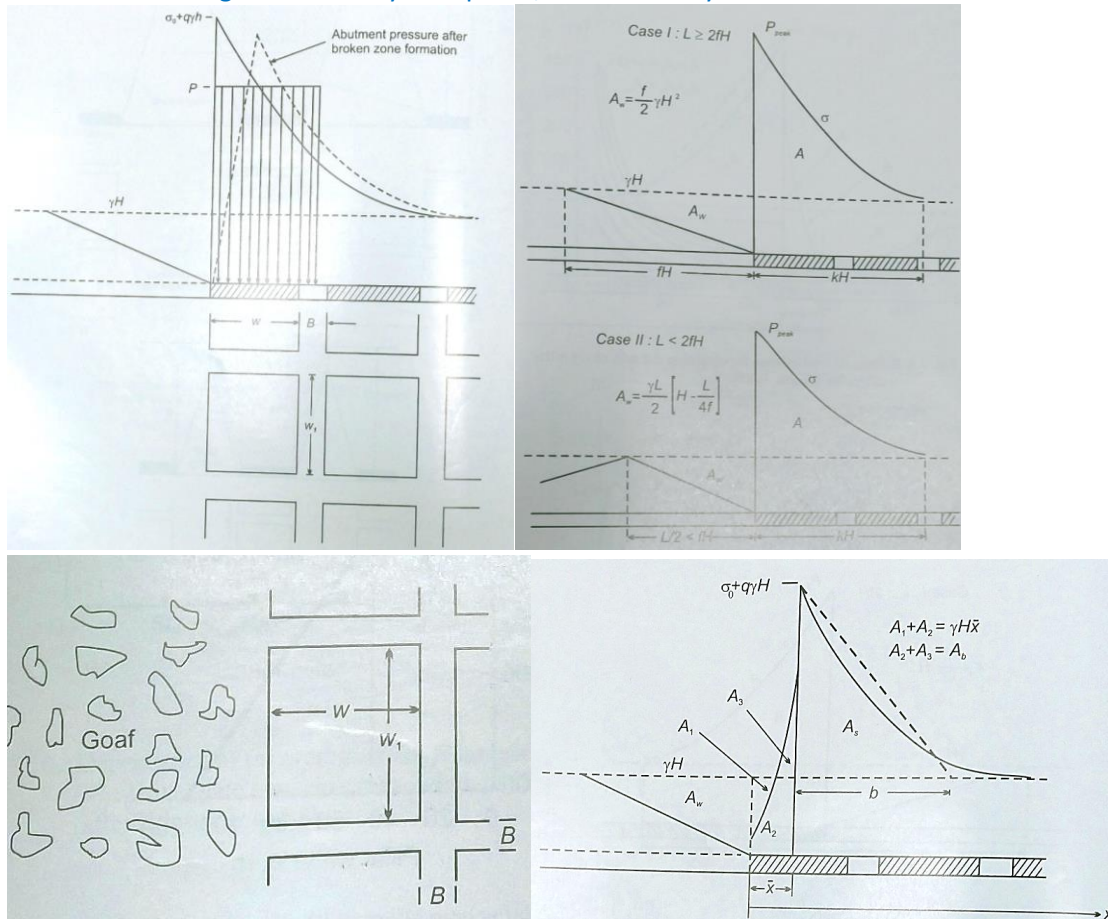
If goaf exists on both sides, the pillars will be subjected to loads (b) and (c) twice during their life period.

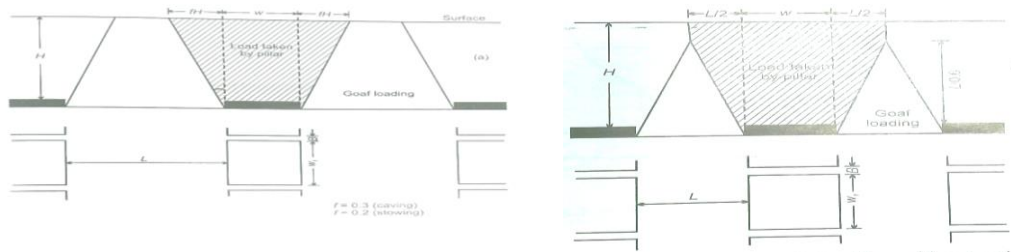
The three-dimensional nature of the problems defies the estimation of load (b) by any analytical means.

However, since the load (b) is only a fraction of load (c), pillars designed to deal with this load are sufficient to ensure their stability during extraction.

The load (a) is estimated using the tributary area concept while load (c) is estimated using Wilson's concept explained above. Wilson analysed roof-to-floor convergence measurements behind caved longwall faces and based on these he proposed a linear rise in the vertical goaf stress starting from zero at the ribside or face to the cover pressure γH at a distance of **0.3–0.4H**. American findings from extensive measurements propose this distance of goaf settlement on an average as **0.38H** (Mark, 1992). In India the value of **0.3H** is taken for normally cavable strata. For hard, difficult to cave strata it is assumed that no caving takes place and the goaf does not offer any support (Sinha, 2001). Some differences in this distance can obviously exist depending on the nature of strata. Some observations have put this distance of **0.2H** for stowed panels. These considerations lead to the distribution of vertical load in the goaf as shown. If the panel width is less than **2fh**, the load in the centre will not reach cover load. If goafs exist on both sides of the pillar chain, the average pillar load could be obtained from Figure. For all these variations of normal caving, stowing and uncavability Table gives the formulae for estimating the load on chain pillars when they have goafs on both sides. Some preliminary calculations using pillar strength Equations and the load formulae of Table will show that Regulation pillars will generally have safety factors less than 2.0.

When long-term stability is required, two rows may have to be left.



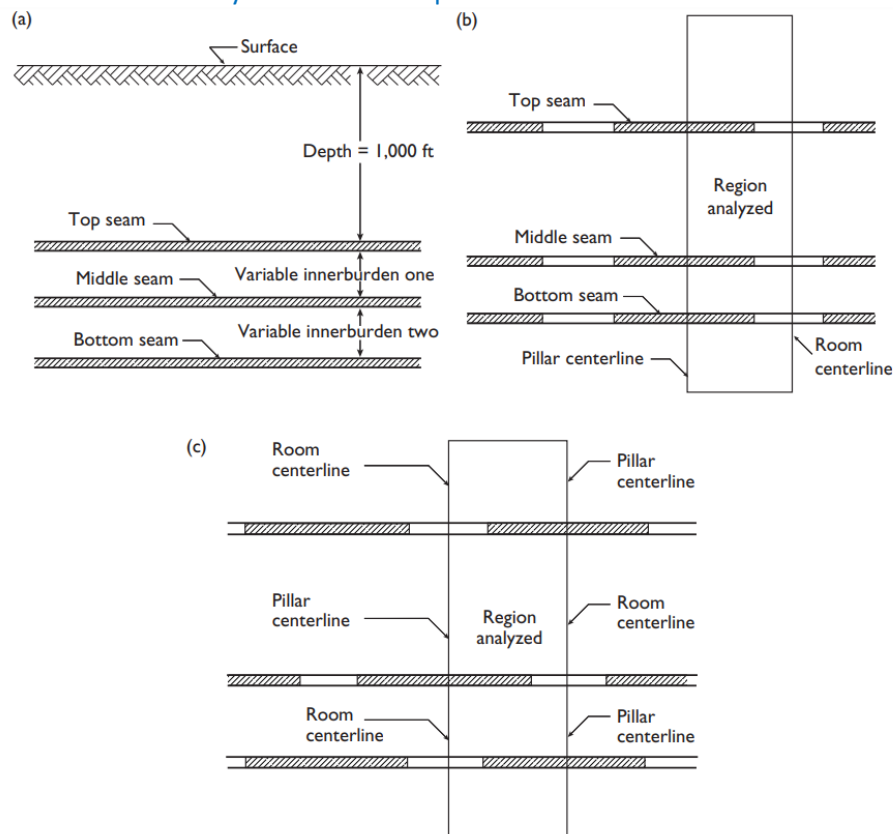


24. Explain the pillar design principles in multi-seam working.

Multiseam mining involves simultaneous or sequential excavation in two or more seams. Columnize the mains and stagger the chains

Columnized main entry pillars: In the columnized case, pillars between entries have the same dimensions and experience about the same average stress, $S_p = S_v \frac{(W_o + W_p)}{W_p}$, when seam separation is small compared with seam depth. Thus, there is no great difference in pillar safety factors compared with single seam tributary pillar analysis. With respect to stress concentration, "columnizing the mains" is advantageous, although, if bed separation occurs in the immediate roof, then there is a lessened advantage.

In the staggered case, the lower entries are offset beneath pillars between the entries above. These lower entries may undermine the pillars between the entries above.



25. What are the geo-mining conditions responsible for occurrences of air blast and rock burst in Bord & Pillar and Longwall Mining Methods?

The following 5 conditions are necessary for rock bursts to occur (Crouch and Fairhurst, 1974).
Depth of cover generally exceeding 300 m.

Strong, stiff roof strata which are not easily cavaled.

Structurally strong coal, not necessarily with a high compressive strength.

Strong floor which cannot heave easily.

Improper mining method producing high stresses.

An air blast can occur under any depth of cover, given the presence of hard strata, a rock burst can occur only at depths generally greater than 300 m and only if the coal itself is burst-prone

26. Mention the different measures to be adopted for control of rock burst and air blast while working under hard massive roof strata.

Control of hard roof strata can be achieved by adopting the following methods in the absence of stowing or filling:

1. Water treatment of the strata
2. Yield pillar technique
3. Roof blasting
4. Goaf edge support measures

These methods are discussed below primarily with air blasts in view, through the first two may have application for rock burst control.

Combating and control of bursts: Volley Firing

Destressing by volley firing has successfully reduced the number of coal bumps or bursts in underground mines. In this method peak stress locations are first determined from drilling data. For relieving stress the high stress locations of a longwall face are then drilled and blasted to destroy the integrity of the coal. The face is then advanced through blasted coal. The blast holes are loaded with 1.4 kg of permissible explosives, stemmed and detonated. The drill pattern consists of a series of 50 mm holes, 4-4.5 m deep, drilled on approximately 1.2 m centres. The hole depth depends on the required daily advance of the face and on the location and magnitude of the stress abutment ahead of the face. The corners of the advancing longwall face require a radial drilling pattern, using combinations of two or three holes angled at 10° and 45° to relieve high stress in both the face and rib areas

Hydraulic fracturing

Hydraulic fracturing requires drilling holes about 30 m deep into the panel rib side and injecting high pressure water into the seam. This process fractures the coal and relieves accumulated stress. Though the method is time consuming, and is not recommended for use on the face, it has been found to be most effective in the roof and coal seam ahead of the longwall face (Haramy and McDonnell, 1988). Fracturing a very large and highly stressed area causes loads to be redistributed and may create burst conditions in the mine.

Auger drilling

Drilling large diameter holes, 150-300 mm, using augers has been in practice in Europe and the US for stress relief in burst-prone longwall faces (Talman and Shrooder, 1958; Haramy et al., 1985; Vinokur and Nevolin, 1972; Brauner, 1988). The holes are placed 4-6 m apart and the length traverses the estimated high stress zone. Such drilling has to be done by remote operation since a burst may occur by the drilling itself.

27. Briefly explain the different direct and indirect methods to estimate bump proneness of a coal seam.

The total energy imparted to the specimen is composed of two components: one is elastic strain energy E_e , which is released immediately upon release of applied stress and the other is permanent strain energy E_p , which is irrecoverable upon release of applied loading.

W_{ET} is simply defined as $W_{ET} = \frac{E_e}{E_p}$

Using this index, liability to bursts is classified as follows:

Degree of bump liability	W_{ET}
Severely liable	> 5
Slightly liable	2-5
Not liable	< 2

According to the Czech method (Zamarski and Franek, 1977) a seam is bump prone if the following two conditions are simultaneously satisfied:

1. Elastic strain limit of coal $\geq 70\%$
2. $K \geq 3.0$

where $K = K_1 + K_2 + K_3$;

$$K_1 = \frac{\text{Depth of cover (m)}}{300};$$

$$K_2 = \frac{\text{Thickness (m) of roof strata of strength} > 700 \text{ kg/cm}^2}{10};$$

$$K_3 = \frac{\text{Average strength of immediate strata (kg/cm}^2\text{)}}{700}$$

The thickness of the immediate strata is taken five times the seam thickness.

The other method, proposed by Brauner of Germany (1988), consists of keeping a cubic coal specimen under total confinement in steel, giving a fixed load on top and drilling a small hole at its side. The resulting stress relaxation with drilling is plotted on a closed-loop servo control machine. If the relaxation curve shows steps, the coal is bump prone. Each of the indices quoted above leads to classifications or criteria for the bursting liability of coal.

However, for complete predictions of rock burst or bump hazard for a specific location in a mine, data on insitu stress concentrations must also be collected. Since direct stress measurements in coal are not recommendable as a routine procedure in mines, a number of indirect detecting methods have been tried which include:

(a) deformation measurements method: Rock deformation measurements such as convergence or convergence rate and pillar deformation have been successfully used to detect burst-prone situations underground at the face or in the entries. In German coal mines (Brauner, 1989), in order to identify a burst prone situation, the ratio of rib side extrusion and roof-to-floor convergence has been found to be a useful index. The case study of Chinakuri No. 1 mine gives the results of vertical pillar deformation measurements which clearly indicate the onset of a burst. The US Bureau of Mines carried out stress measurement in chain pillars left between two longwall panels extracted one after the other in a burst-prone seam (Iannachione, 1990). Convergence was also measured in adjacent entries (gateroads). Similar results of measurements will doubtless be obtained in a mine which is not burst-prone. So, the obvious requirement is that the seam conditions must be burst-prone for such underground observations to be meaningful for prediction or detection of bursts.

(b) drilling yield method: The drilling yield method is widely used throughout the world to locate zones of abnormal or high stress concentration (Kidybinski, 1981; Neyman et al., 1972; Noack et al., 1983; Brauner, 1989). This method, being used in the USSR since the 1950's, was modified and adapted to local geological conditions in the early 1960's in a few of the European coalfields. The method was later introduced to U.S. coal mines and was favorably accepted (Haramy and McDonnel, 1988). This method involves drilling holes into the seam and recording the volume of drill cuttings obtained from each metre of a given hole depth. A certain volume of cuttings can be expected from a drill hole that has a known diameter and length. If the actual volume of cuttings generated exceeds the volume of the hole by a significant amount, the zone around that particular hole is determined to be highly stressed. The critical values for drilling yields applied in Germany (Noack et al., 1983) are 90 l/m for 140 mm diameter boreholes; 50 l/m for 95 mm diameter holes; 8 l/m for 50 mm boreholes and 6 l/m for 42 mm boreholes. Above these values, the stress concentration is dangerous and a critical state of burst hazard is assumed. It is to be noted that such high yields of dust must be accompanied by drill jamming, grating etc. Studies carried out in Germany and Poland provided a basis for stating a relationship between burst potential determined from drilling yield and extracted seam thickness. If a highly stressed zone is detected at a distance of more than 3.4 times the seam thickness from the rib side, there is believed to be no danger but when the zone is less than 1.5 times the seam thickness away an extremely hazardous state exists and men should be withdrawn.

(c) microseismic method: The microseismic method, also known as the acoustic emissions (AE) method, was developed by the U.S. Bureau of Mines in the late 1930's (Obert and Duvall, 1945; 1967). It is a geophysical approach that detects subaudible rock noise (microseisms) associated with yielding or fracturing of the rock. Geophones translate the rock noises from acoustic waves, which may be undetectable by the unaided ear, to electrical analogue signals. Laboratory studies of rock indicate increased noise rate with increased stress, the rate of increase becoming pronounced as the ultimate stress is approached. A pattern of increased activity followed by a quiet period immediately prior to a bump has been noted by several investigators, but this behaviour has not been consistently reproduced in the laboratory.

Section C: Numerical Type Questions – 25 marks

Answer all the questions and marks are assigned against each question

28. Design a Bord & Pillar panel (number of pillars along length and width) for assumed pillar size of 30 m x 30 m developed in a 3 m thick coal seam with incubation period of 12 months to be extracted using Continuous Miner Technology in Bord and Pillar Mining Method. Assume pillar recovery during depillaring as 70%, working days of 300 in a year and daily desired production of 2000 tons from a Continuous Miner panel.

(3 marks)

Pillar size – 30 m x 30 m

Coal seam height – 3 m

Average daily production – 2000 t

Incubation period – 12 months

Working days – 300 days

Recovery from a coal pillar – 70%

Extractable coal in a panel with pillars 'n' = 30 m x 30 m x 3 m x 1.3 t/m³ x 0.70 x n

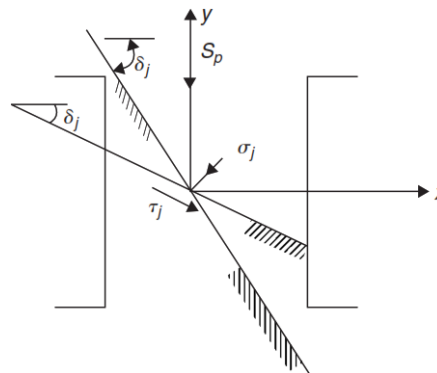
30 m x 30 m x 4 m x 1.3 t/m³ x 0.70 x n = 2000 x 300

$$n = \frac{2000 \times 300}{30 \times 30 \times 3 \times 1.3 \times 0.70} = 240$$

Along width = 3-5 pillars and length = 48-80 pillars considering the easy caveability in straight line of extraction in Bord and Pillar working using CM Technology.

29. Suppose that the pillar stress is 29 MPa in a Bord and Pillar mine and that a joint set is present with an average bearing of N45°W (as shown in Figure). If the joints obey a Mohr–Coulomb slip criterion and have a friction angle of 35° and a cohesion of 4 MPa, what range of joint dips is unsafe with respect to joint slip?

(4 marks)

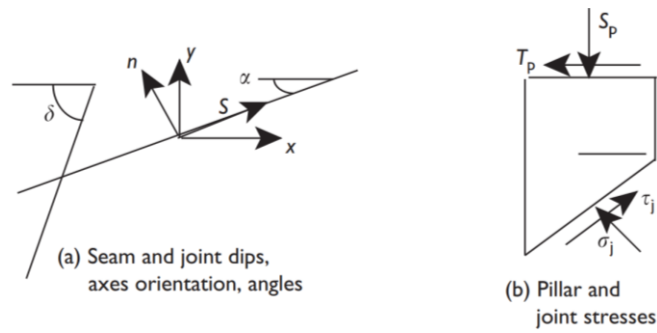


$$\begin{aligned} \left(\frac{S_p}{2}\right) \sin(2\delta_j - \phi_j) &> \left(\frac{S_p}{2}\right) \sin(\phi_j) + c_j \cos(\phi_j) \\ \sin(2\delta_j - \phi_j) &< \sin(\phi_j) + \frac{2c_j}{S_p} \cos(\phi_j) = \sin(35^\circ) + \frac{2 \times 4}{29} \cos(35^\circ) = 0.80 \\ \sin(2\delta_j - \phi_j) &< 0.80 \\ 2\delta_j - \phi_j &> 53 \\ \delta_j &> 44 \\ 80.5 &> \delta_j > 44 \end{aligned}$$

30. A room and pillar operation is planned at a depth of 964 ft in a seam dipping 13°. An extraction percentage of 50% is anticipated. The ratio of pre-excitation horizontal to vertical stress is estimated to be one-fourth. A joint set is present that dips 60° in the same direction as the seam. Rock and joint properties are given in the table. Determine if there is a threat to pillar stability. (Use $\gamma = 1$ psi/ft)

Property material	Cohesion (psi)	Friction angle (°)
Rock	2300	35
Joint	23	25

(8 marks)



A threat to pillar stability exists if either the pillar safety factor or joint safety factor is less than one. Pillar strength and joint properties are given, while pillar stress may be determined from extraction ratio formulas. Joint stresses may be determined from equilibrium requirements. With reference to the sketch, the pre-excitation normal and shear stresses are

$$S_n = \frac{S_x + S_y}{2} - \frac{S_x - S_y}{2} \cos 2\alpha + T_{xy} \sin 2\alpha = \frac{S_v(k+1)}{2} - \frac{S_v(k-1)}{2} \cos 26 + 0 \sin 26$$

$$= \frac{964(\frac{1}{4}+1)}{2} - \frac{964(\frac{1}{4}-1)}{2} \cos 26 + 0 \sin 26 = 927 \text{ psi}$$

$$T_s = -\left(\frac{S_x - S_y}{2}\right) \sin 2\alpha + T_{yz} \cos 2\alpha = -\frac{964(\frac{1}{4}-1)}{2} \sin 26 + 0 \cos 26 = 158 \text{ psi}$$

$$\text{The pillar stresses are } S_p = \frac{S_v}{1-R} = \frac{927}{1-\frac{1}{2}} = 1854 \text{ psi and } T_p = \frac{T_s}{1-R} = \frac{158}{1-\frac{1}{2}} = 316 \text{ psi}$$

$$\text{The pillar factor of safety is given by } FS = \frac{\frac{2c \cos \phi}{1 - \sin(\phi - \beta)}}{\sqrt{(S_p)^2 + (T_p)^2}} = \frac{\frac{2 \times 2300 \cos 35}{1 - \sin(35 - 9.7)}}{\sqrt{(1854)^2 + (316)^2}} = 3.5$$

Thus, the pillar safety factor indicates no threat of failure. The joint stresses are

$$\sigma_j = \frac{S_p}{2} [1 + \cos(2(\delta - \alpha))] - \frac{T_p}{2} \sin(2(\delta - \alpha))$$

$$= \frac{1854}{2} [1 + \cos 94] - \frac{316}{2} \sin 94 = 705 \text{ psi}$$

$$\tau_j = \frac{1854}{2} \sin 94 - \frac{T_p}{2} [1 + \cos(94)] = 1072 \text{ psi}$$

$$FS_j = \frac{\sigma_j \tan(\phi_j) + c_j}{\tau_j} = \frac{705 \tan(25) + 23}{1072} = 0.328$$

Thus, joint failure poses a threat to pillar stability in this excavation plan.

31. Bord and Pillar Mining Method using Continuous Miner Technology is being implemented for extraction of 6 m thick coal seam at a depth of cover of 300 m with Rock Mass Rating (RMR) of 50 and Young's Modulus of roof of 3 GPa. The size of developed coal pillar is 30 m x 30 m and gallery width and height of 6 m and 4.5 m respectively. Use the appropriate empirical formulations to design the following geotechnical elements with appropriate diagram (wherever required)

(i) Cut-Out Distance (COD) (0.5 mark)

$$S = 14.61 + 1.98E - 2.12W = 14.61 + 1.98 \times 3 - 2.12 \times 6 = 7.83 \text{ m} \approx 8 \text{ m}$$

(ii) Extent of Pillar Spalling (0.5 mark)

$$x_0 = 0.00492hH = 0.00492 \times 4.5 \times 300 = 6.642 \text{ m}$$

(iii) Support design in galleries and junctions of 6.0 m (width) x 4.5 m (height) (1 mark)

$$\text{Rock load in roadways} = B\gamma(1.7 - 0.037RMR + 0.0002RMR^2)$$

$$= 6 \times 2.5 \times (1.7 - 0.037 \times 50 + 0.0002 \times 50^2) = 5.25 \text{ tons/m}^2$$

Let number of bolts = n; grouting strength is 20 tons for each bolt and grid pattern as 1.2 m x 1.2 m

$$\text{Support Resistance } (S_r) = \frac{\text{Number of support} \times \text{Capacity of each support}}{\text{Area supported by one row of supports}} = \frac{n \times 20}{6 \times 1.2}$$

$$\text{Support safety factor} = \frac{\text{Support Resistance}}{\text{Rock Load}} = \frac{\frac{n \times 20}{6 \times 1.2}}{5.25} = 2$$

$$n = \frac{5.25 \times 2 \times 6 \times 1.2}{20} = 3.78 \approx 4 \text{ bolts}$$

$$\text{Rock load in junctions} = 5B^{0.3}\gamma\left(1 - \frac{RMR}{100}\right)^2 = 5 * 6^{0.3} * 2.5\left(1 - \frac{50}{100}\right)^2$$

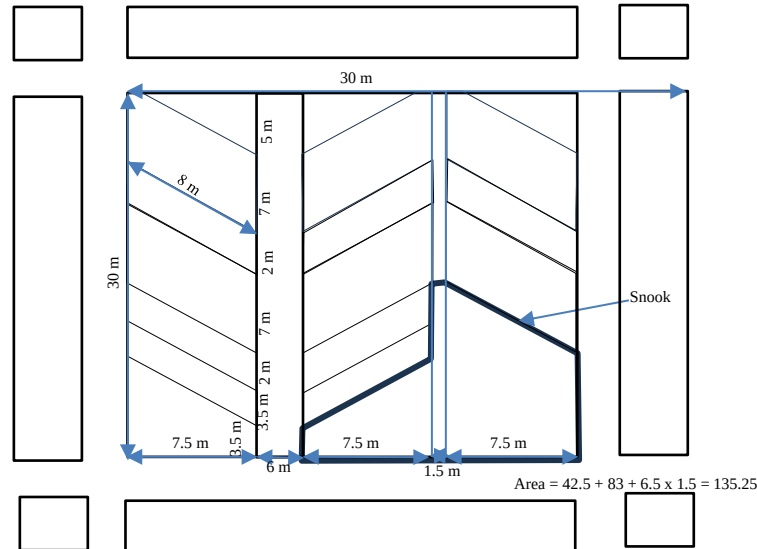
$$= 5.35 \text{ tons/m}^2$$

$$n = \frac{5.35 * 3 * 6 * 1}{20} \approx 5 \text{ bolts}$$

(iv) Manner of Pillar Extraction based on COD

(2 marks)

Since COD = 8m and pillar dimension is 30 m x 30 m, therefore, each fender width is 7.5 m. Combination of Christmas Tree and Split and slicing. Therefore, Christmas tree method extraction from split gallery of 6 m and split and slicing method of extraction from original galleries.



(v) Rib/Snook size/area with slice dimension

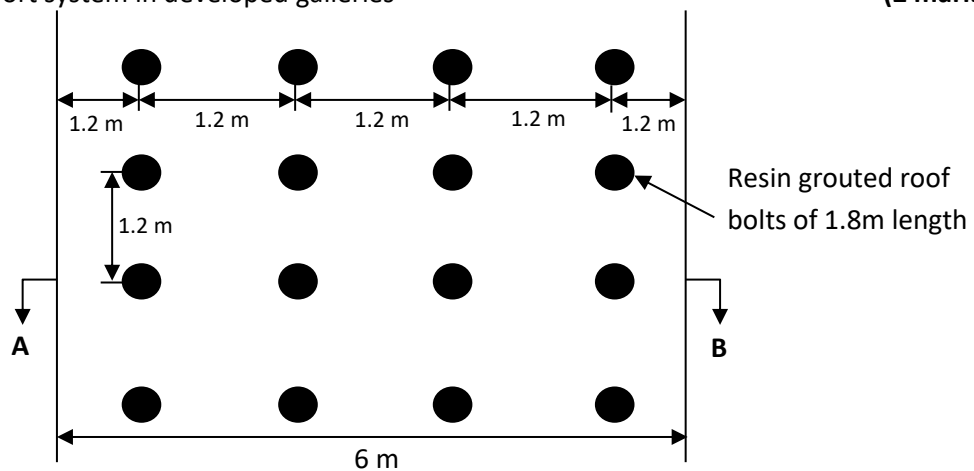
(2 marks)

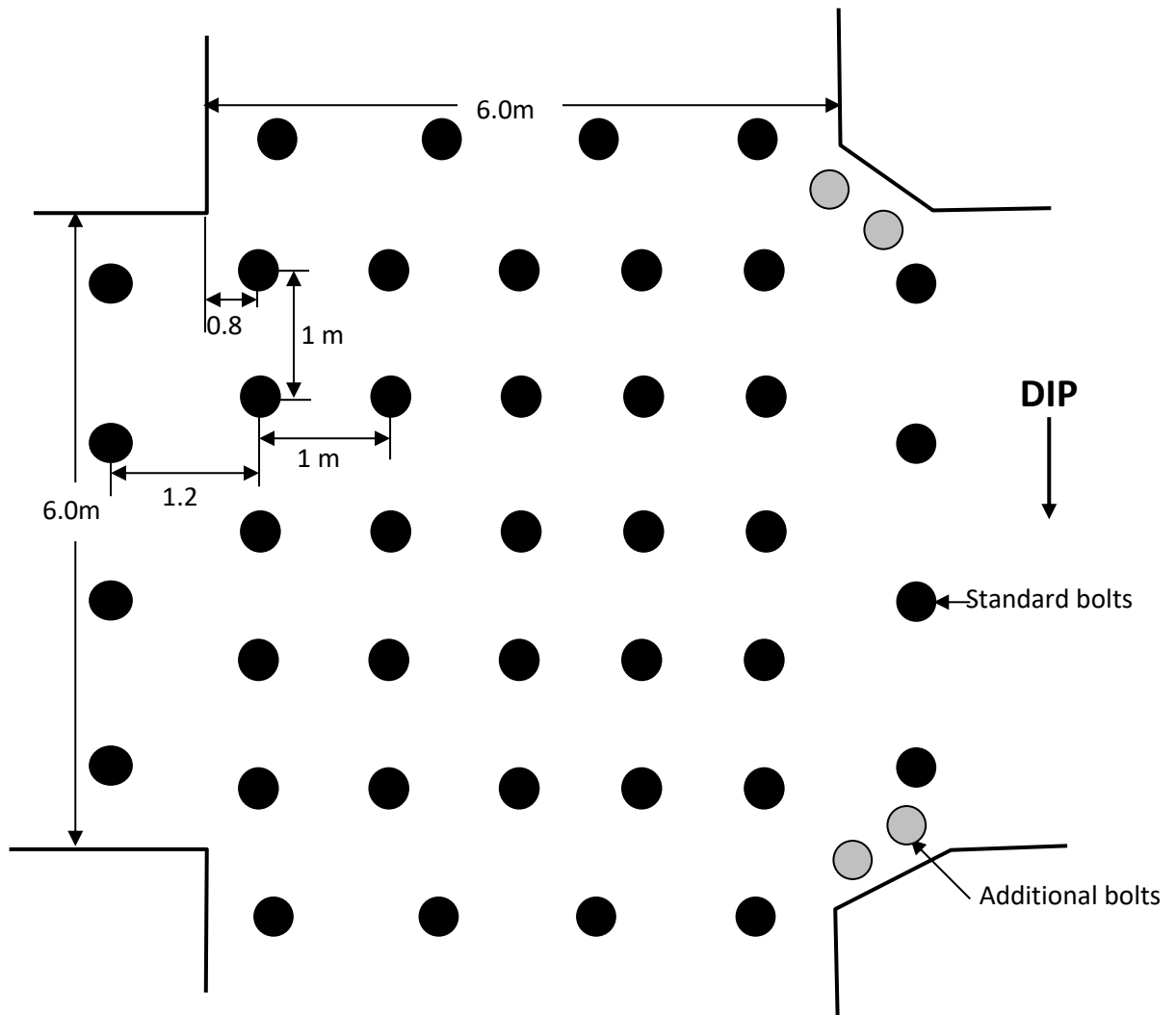
$$A = 0.16H^{0.89}R^{0.14}h^{0.31} = 0.16 * 300^{0.89} * 50^{0.14} * 4.5^{0.31} = 71 \text{ m}^2$$

$$w_e = 0.18H^{0.15}R^{0.10}h^{-0.03}A^{0.60} = 0.18 * 300^{0.15} * 50^{0.10} * 4.5^{-0.03} * 62^{0.60} = 7.7 \text{ m}$$

(vi) Support system in developed galleries

(2 marks)





Proposed support system at junctions

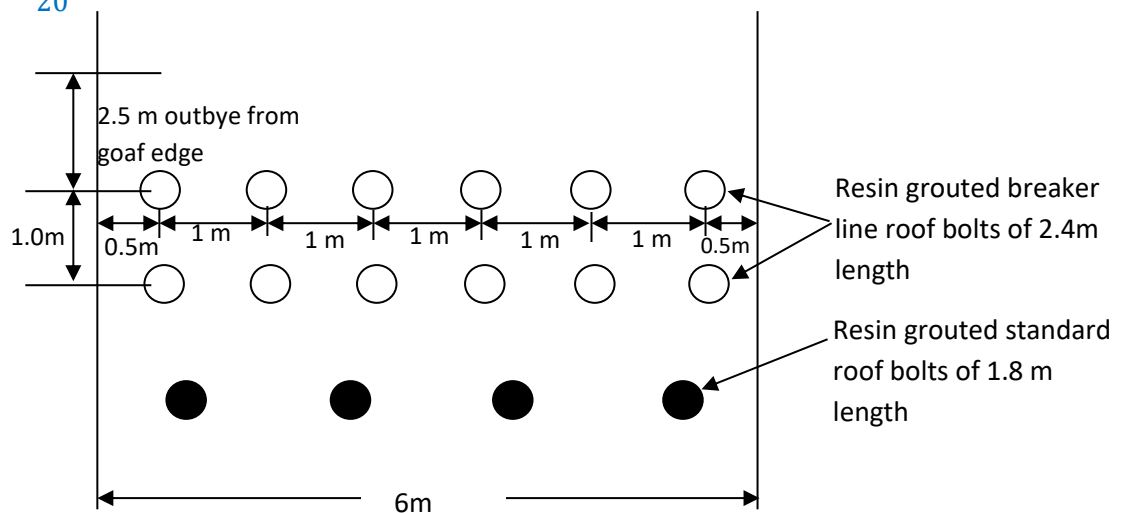
(vii) Location and grid pattern of roof bolts-based goaf edge support (2 marks)

For 2 m out-bye from goaf edge: $RLH = 115.22 H^{0.12} R^{-1.20} = 115.22 * 300^{0.12} * 50^{-1.20} = 2.1 \text{ m}$

Rock Load = $2.5 * 2.1 = 5.25 \text{ tons/m}^2$

$$\text{Support safety factor} = \frac{\text{Support Resistance}}{\text{Rock Load}} = \frac{\frac{n \cdot 20}{6 \cdot 0.8}}{5.25} = 4$$

$$n = \frac{5.25 * 4 * 6 * 1}{20} = 6.3 \approx 6 \text{ bolts}$$



Proposed breaker line support system to be installed

Note: Formulas to be used:

For Rib/Snook

$$A = 0.16H^{0.89}R^{0.14}h^{0.31}$$

$$\frac{w_e}{h} = 0.18H^{0.15}R^{0.10}h^{-1.03}A^{0.60}$$

$$w_e = 0.18H^{0.15}R^{0.10}h^{-0.03}A^{0.60}$$

For Support system in developed galleries

$$\text{Rock load in roadways} = B\gamma(1.7 - 0.037RMR + 0.0002RMR^2)$$

$$\text{Rock load in junctions} = 5B^{0.3}\gamma\left(1 - \frac{RMR}{100}\right)^2$$

For RBBLS

$$\text{For 0 m out-bye from goaf edge: } RLH = 11.67 H^{0.58} R^{-1.14}$$

$$\text{For 1 m out-bye from goaf edge: } RLH = 66.32 H^{0.31} R^{-1.26}$$

$$\text{For 2 m out-bye from goaf edge: } RLH = 115.22 H^{0.12} R^{-1.20}$$

For Pillar Spalling

$$x_0 = 0.00492hH$$

For Cut-out distance

$$S = 14.61 + 1.98E - 2.12W$$